CHAPTER FIVE

BASIC PLANAR KINEMATICS OF A RIGID BODY

Learning Objectives

After completing this chapter, students should be able to do the following:

- Classify the various types of rigid-body planar motion.
- Investigate rigid-body translation and angular motion about a fixed axis.

5.1 Planar Rigid-Body Motion

This study is important for the design of gears, cams, and mechanisms used for many mechanical operations. Once the kinematics is thoroughly understood, then we can apply the equations of motion, which relate the forces on the body to the body's motion.

The planar motion of a body occurs when all the particles of a rigid body move along paths which are equidistant from a fixed plane. There are three major types of rigid body planar motion, in order of increasing complexity.

Fig. 5.1: Examples of types of rigid body planar motion

1. **Translation:** This type of motion occurs when a line in the body remains parallel to its original orientation throughout the motion. When the paths of motion for any two points on the body are parallel lines, the motion is called rectilinear translation, Fig. 5.1 (a). If the paths of motion are along curved lines which are equidistant, the motion is called curvilinear translation, Fig. 5.1 (b).

2. **Rotation about a fixed axis:** When a rigid body rotates about a fixed axis, all the particles of the body, except those which lie on the axis of rotation, move along circular paths, Fig. 5.1 (c).

3. **General plane motion:** When a body is subjected to general plane motion, it undergoes a combination of translation and rotation, Fig. 5.1 (d). The translation occurs within a reference plane, and the rotation occurs about an axis perpendicular to the reference plane.

5.2 Translation

Consider a rigid body which is subjected to either rectilinear or curvilinear translation in the x-y plane, Fig. 5.2.

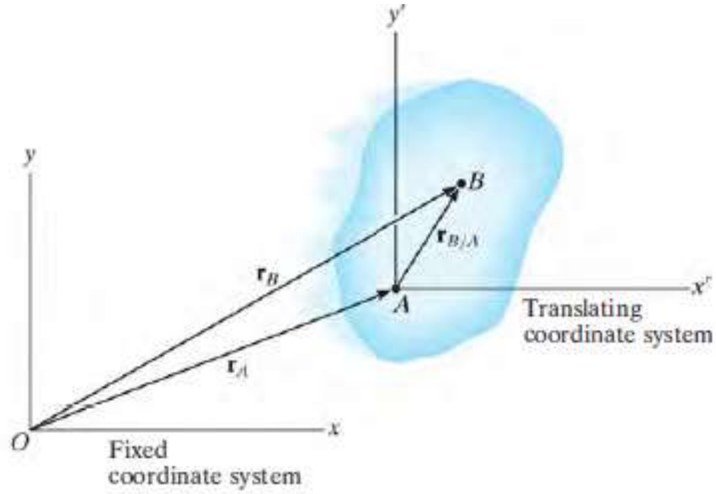

Fig. 5.2

5.2.1 Position

The locations of points A and B on the body are defined with respect to fixed x, y reference frame using *position vectors* $\boldsymbol{r_A}$ and $\boldsymbol{r_B}$. The translating x', y' coordinate system is fixed in the body and has its origin at A, hereafter referred to as the base point. The position of B with respect to A is denoted by the *relative-position vector* $\boldsymbol{r_{B/A}}$ ("$\boldsymbol{r}$ of B with respect to A"). By vector addition,

$$\boldsymbol{r_B} = \boldsymbol{r_A} + \boldsymbol{r_{B/A}} \tag{5.1}$$

5.2.2 Velocity

A relation between the instantaneous velocities of A and B is obtained by taking the time derivative of this equation, which yields $\boldsymbol{v_B} = \boldsymbol{v_A} + d\boldsymbol{r_{B/A}}/dt$. Here $\boldsymbol{v_A}$ and $\boldsymbol{v_B}$ denote *absolute velocities* since these vectors are measured with respect to the x, y axes. The term $d\boldsymbol{r_{B/A}}/dt = 0$, since the *magnitude* of $\boldsymbol{r_{B/A}}$ is constant by definition of a rigid body, and because the body is translating the *direction* of $\boldsymbol{r_{B/A}}$ is also constant. Therefore,

$$\boldsymbol{v_B} = \boldsymbol{v_A}$$

5.2.3 Acceleration

Taking the time derivative of the velocity equation yields a similar relationship between the instantaneous accelerations of A and B:

$\boldsymbol{a_B} = \boldsymbol{a_A}$

The above two equations indicate that all points in a rigid body subjected to either rectilinear or curvilinear translation move with the same velocity and acceleration. As a result, the kinematics of particle motion, discussed in Chapters 1 and 2, can also be used to specify the kinematics of points located in a translating rigid body.

5.3 Rotation About a Fixed Axis

Consider the angular motion of a body about a specified axis. When a body rotates about a fixed axis, any point P located in the body travels along a circular path, Fig. 5.3.

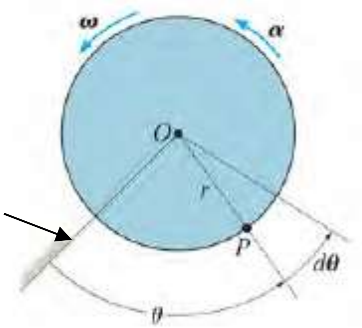

Fig. 5.3

5.3.1 Angular Position

At the instant shown, the angular position of P is defined by the angle θ, measured from a fixed reference line to r.

5.3.2 Angular Displacement

The change in the angular position, which can be measured as a differential $d\theta$, is called the *angular displacement*. This vector has a *magnitud*e of $d\theta$, measured in degrees, radians, or revolutions, where $1\, rev = 2\pi\, rad$. Since motion is about a *fixed axis*, the .direction of $d\theta$ is always along this axis. Specifically, the direction is determined by the right-hand rule; that is, the fingers of the right hand are curled with the sense of rotation, so that in this case the thumb, or $d\theta$, points upward.

5.3.3 Angular Velocity

The time rate of change in the angular position is called the angular velocity ω (omega). Since $d\theta$ occurs during an instant of time , then,

$$\omega = \frac{d\theta}{dt} \quad (+ACW) \tag{5.2}$$

5.3.4 Angular Acceleration

The angular acceleration a (alpha) measures the time rate of change of the angular velocity. The magnitude of this vector is

$$\alpha = \frac{d\omega}{dt} \quad (+ACW) \tag{5.3}$$

Using Eq. (5.2), it is also possible to express α as

$$\alpha = \frac{d^2\theta}{dt^2} \quad (+ACW) \tag{5.4}$$

By eliminating dt from Eqs. 5.2 and 5.3, we obtain a differential relation between the angular acceleration, angular velocity, and angular displacement, namely,

$$\alpha d\theta = \omega d\omega \ (+ACW) \tag{5.5}$$

The similarity between the differential relations for angular motion and those developed for rectilinear motion of a particle $v = \frac{ds}{dt}$, $a = \frac{dv}{dt}$, and $ads = vdv$) should be apparent.

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5.3.5 Constant Angular Acceleration

If the angular acceleration of the body is constant, $\alpha = \alpha_c$, then Eqs. 5.2, 5.3, and 5.5, when integrated, yield a set of formulas which relate the body's angular velocity, angular position, and time. These equations are similar to Eqs. 1.1 to 1.3 used for rectilinear motion. The results are presented for constant angular acceleration.

$$(+ACW) \quad \omega = \omega_0 + \alpha_c t \tag{5.6}$$

$$(+ACW) \quad \theta = \theta_0 + \omega_0 t + \frac{1}{2}\alpha_c t^2 \tag{5.7}$$

$$(+ACW) \quad \omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0) \tag{5.8}$$

Here θ_0 and ω_0 are the initial values of the body's angular position and angular velocity, respectively.

5.4 Motion Analysis of Point P

As the rigid body in Fig. 5.3 rotates, point P travels along a circular path of radius r with center at point O.

5.4.1 Position and Displacement

The position of P is defined by the position vector $\boldsymbol{r}$, which extends from 0 to P. If the body rotates $d\theta$ then the displacement of P will be

$$ds = rd\theta \tag{5.9}$$

5.4.2 Velocity

The velocity of P has a magnitude which can be found by dividing $ds = rd\theta$ by dt so that

$$v = \omega r \tag{5.10}$$

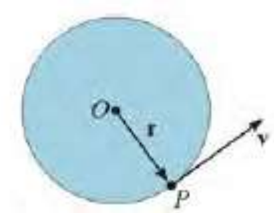


Fig. 5.4

As shown in Fig. 5.4, the direction of $\boldsymbol{v}$ is tangent to the circular path.

5.4.3 Acceleration

The acceleration of P can be expressed in terms of its normal and tangential components as indicated in Fig. 5.5.

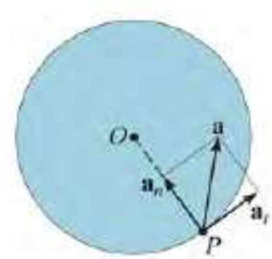


Fig. 5.5

Since $a_t = dv/dt$ and $a_n = v^2/\rho$, where $\rho = r$, $v = \omega r$, and $\alpha = d\omega/dt$, we have

$$a_t = \alpha r \tag{5.9}$$

$$a_n = \omega^2 r \tag{5.10}$$

Hence,

$$\boldsymbol{a} = \boldsymbol{a_t} + \boldsymbol{a_n} = \boldsymbol{\alpha} \times \boldsymbol{r} - \omega^2 \boldsymbol{r} \tag{5.11}$$

Since $\boldsymbol{a_t}$ and $\boldsymbol{a_n}$ are perpendicular to one another, if needed the magnitude of acceleration can be determined from the Pythagorean theorem; namely,

$$a = \sqrt{a_n^2 + a_t^2} \tag{5.12}$$

Note: If the geometry of the problem is difficult to visualize, the following vector equations should be used:

$$\boldsymbol{v} = \boldsymbol{\omega} \times \boldsymbol{r}_P = \boldsymbol{\omega} \times \boldsymbol{r} \tag{5.13}$$

(5.14)

(5.15)

$\boldsymbol{a_t} = \boldsymbol{\alpha} \times \boldsymbol{r}_P = \boldsymbol{\alpha} \times \boldsymbol{r}$

$\boldsymbol{a_n} = \boldsymbol{\omega} \times (\boldsymbol{\omega} \times \boldsymbol{r}_P) = -\omega^2 \boldsymbol{r}$

5.5 Absolute Motion Analysis

A body subjected to general plane motion undergoes a simultaneous translation and rotation. If the body is represented by a thin slab, the slab translates in the plane of the slab and rotates about an axis perpendicular to this plane. The motion can be completely specified by knowing both the angular rotation of a line fixed in the body and the motion of a point on the body. One way to relate these motions is to use a rectilinear position coordinate s to locate the point along its path and an angular position coordinate θ to specify the orientation of the line. The two coordinates are then related using the geometry of the problem. By direct application of the time-differential equations $v = ds/dt, a = dv/dt$, $\omega = d\theta/dt$ and $\alpha = d\omega/dt$, the motion of the point and the angular motion of the line can then be related. In some cases, this same procedure may be used to relate the motion of one body, undergoing either rotation about a fixed axis or translation, to that of a connected body undergoing general plane motion.

Example 5.1

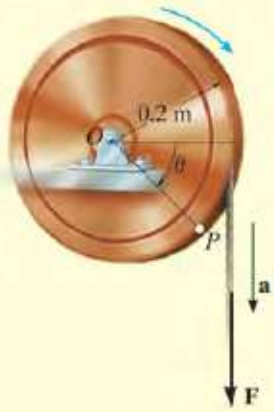


Fig. 16–5

A cord is wrapped around a wheel in Fig. 16–5, which is initially at rest when $\theta = 0$. If a force is applied to the cord and gives it an acceleration $a = (4t)\ \text{m/s}^2$, where t is in seconds, determine, as a function of time, (a) the angular velocity of the wheel, and (b) the angular position of line OP in radians.

SOLUTION

Part (a). The wheel is subjected to rotation about a fixed axis passing through point O. Thus, point P on the wheel has motion about a circular path, and the acceleration of this point has *both* tangential and normal components. The tangential component is $(a_P)_t = (4t)\ \text{m/s}^2$, since the cord is wrapped around the wheel and moves *tangent* to it. Hence the angular acceleration of the wheel is

$$(\circlearrowright +) \qquad (a_P)_t = \alpha r$$
$$(4t)\ \text{m/s}^2 = \alpha(0.2\ \text{m})$$
$$\alpha = (20t)\ \text{rad/s}^2 \circlearrowright$$

Using this result, the wheel's angular velocity ω can now be determined from $\alpha = d\omega/dt$, since this equation relates α, t, and ω. Integrating, with the initial condition that $\omega = 0$ when $t = 0$, yields

$$(\circlearrowright +) \qquad \alpha = \frac{d\omega}{dt} = (20t)\ \text{rad/s}^2$$
$$\int_0^{\omega} d\omega = \int_0^{t} 20t\ dt$$
$$\omega = 10t^2\ \text{rad/s} \circlearrowright \qquad \textit{Ans.}$$

Part (b). Using this result, the angular position θ of OP can be found from $\omega = d\theta/dt$, since this equation relates θ, ω, and t. Integrating, with the initial condition $\theta = 0$ when $t = 0$, we have

$$(\circlearrowright +) \qquad \frac{d\theta}{dt} = \omega = (10t^2)\ \text{rad/s}$$
$$\int_0^{\theta} d\theta = \int_0^{t} 10t^2\ dt$$
$$\theta = 3.33t^3\ \text{rad} \qquad \textit{Ans.}$$

NOTE: We cannot use the equation of constant angular acceleration, since α is a function of time.

Example 5.2

The motor shown in the photo is used to turn a wheel and attached blower contained within the housing. The details of the design are shown in Fig. 16–6*a*. If the pulley A connected to the motor begins to rotate from rest with a constant angular acceleration of $\alpha_A = 2\text{ rad/s}^2$, determine the magnitudes of the velocity and acceleration of point P on the wheel, after the pulley has turned two revolutions. Assume the transmission belt does not slip on the pulley and wheel.

SOLUTION

Angular Motion. First we will convert the two revolutions to radians. Since there are 2π rad in one revolution, then

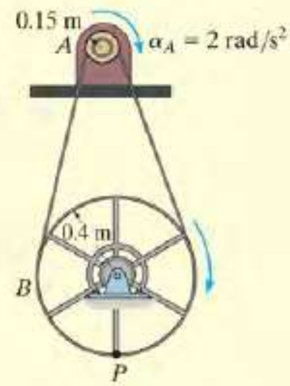
$$\theta_A = 2\text{ rev}\left(\frac{2\pi\text{ rad}}{1\text{ rev}}\right) = 12.57\text{ rad}$$

Since α_A is constant, the angular velocity of pulley A is therefore

$$(\circlearrowright +)\qquad \omega^2 = \omega_0^2 + 2\alpha_c(\theta - \theta_0)$$

$$\omega_A^2 = 0 + 2(2\text{ rad/s}^2)(12.57\text{ rad} - 0)$$

$$\omega_A = 7.090\text{ rad/s}$$



(a)

The belt has the same speed and tangential component of acceleration as it passes over the pulley and wheel. Thus,

$$v = \omega_A r_A = \omega_B r_B;\ 7.090\text{ rad/s }(0.15\text{ m}) = \omega_B(0.4\text{ m})$$

$$\omega_B = 2.659\text{ rad/s}$$

$$a_t = \alpha_A r_A = \alpha_B r_B;\ 2\text{ rad/s}^2\,(0.15\text{ m}) = \alpha_B(0.4\text{ m})$$

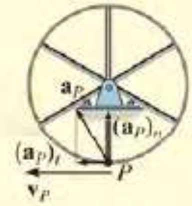
$$\alpha_B = 0.750\text{ rad/s}^2$$

Motion of P. As shown on the kinematic diagram in Fig. 16–6*b*, we have

$$v_P = \omega_B r_B = 2.659\text{ rad/s }(0.4\text{ m}) = 1.06\text{ m/s}\qquad \textit{Ans.}$$

$$(a_P)_t = \alpha_B r_B = 0.750\text{ rad/s}^2\,(0.4\text{ m}) = 0.3\text{ m/s}^2$$

$$(a_P)_n = \omega_B^2 r_B = (2.659\text{ rad/s})^2(0.4\text{ m}) = 2.827\text{ m/s}^2$$



(b)

Thus

$$a_P = \sqrt{(0.3\text{ m/s}^2)^2 + (2.827\text{ m/s}^2)^2} = 2.84\text{ m/s}^2\qquad \textit{Ans.}$$

Fig. 16–6

Example 5.3

The end of rod R shown in Fig. 16–7 maintains contact with the cam by means of a spring. If the cam rotates about an axis passing through point O with an angular acceleration $\boldsymbol{\alpha}$ and angular velocity $\boldsymbol{\omega}$, determine the velocity and acceleration of the rod when the cam is in the arbitrary position θ.

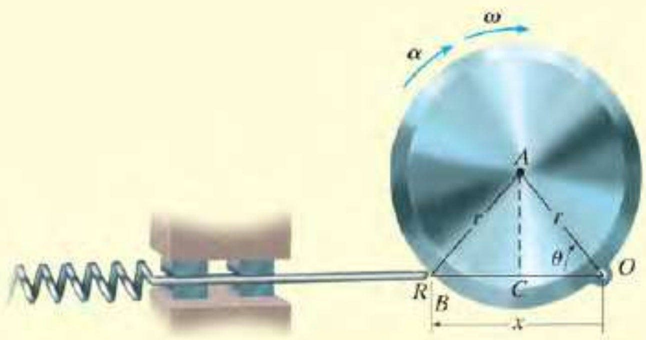


Fig. 16–7

SOLUTION

Position Coordinate Equation. Coordinates θ and x are chosen in order to relate the *rotational motion* of the line segment OA on the cam to the *rectilinear translation* of the rod. These coordinates are measured from the *fixed point O* and can be related to each other using trigonometry. Since $OC = CB = r\cos\theta$, Fig. 16–7, then

$$x = 2r\cos\theta$$

Time Derivatives. Using the chain rule of calculus, we have

$$\frac{dx}{dt} = -2r(\sin\theta)\frac{d\theta}{dt}$$

$$v = -2r\omega\sin\theta \qquad \textit{Ans.}$$

$$\frac{dv}{dt} = -2r\left(\frac{d\omega}{dt}\right)\sin\theta - 2r\omega(\cos\theta)\frac{d\theta}{dt}$$

$$a = -2r(\alpha\sin\theta + \omega^2\cos\theta) \qquad \textit{Ans.}$$

NOTE: The negative signs indicate that v and a are opposite to the direction of positive x. This seems reasonable when you visualize the motion.

Example 5.4

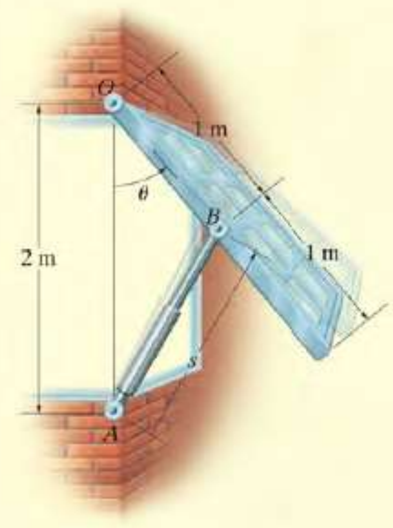


Fig. 16–9

The large window in Fig. 16–9 is opened using a hydraulic cylinder AB. If the cylinder extends at a constant rate of 0.5 m/s, determine the angular velocity and angular acceleration of the window at the instant $\theta = 30°$.

SOLUTION

Position Coordinate Equation. The angular motion of the window can be obtained using the coordinate θ, whereas the extension or motion *along the hydraulic cylinder* is defined using a coordinate s, which measures its length from the fixed point A to the moving point B. These coordinates can be related using the law of cosines, namely,

$$s^2 = (2\text{ m})^2 + (1\text{ m})^2 - 2(2\text{ m})(1\text{ m})\cos\theta$$

$$s^2 = 5 - 4\cos\theta \qquad (1)$$

When $\theta = 30°$,

$$s = 1.239\text{ m}$$

Time Derivatives. Taking the time derivatives of Eq. 1, we have

$$2s\frac{ds}{dt} = 0 - 4(-\sin\theta)\frac{d\theta}{dt}$$

$$s(v_s) = 2(\sin\theta)\omega \qquad (2)$$

Since $v_s = 0.5$ m/s, then at $\theta = 30°$,

$$(1.239\text{ m})(0.5\text{ m/s}) = 2\sin 30°\omega$$

$$\omega = 0.6197\text{ rad/s} = 0.620\text{ rad/s} \qquad \textit{Ans.}$$

Taking the time derivative of Eq. 2 yields

$$\frac{ds}{dt}v_s + s\frac{dv_s}{dt} = 2(\cos\theta)\frac{d\theta}{dt}\omega + 2(\sin\theta)\frac{d\omega}{dt}$$

$$v_s^2 + sa_s = 2(\cos\theta)\omega^2 + 2(\sin\theta)\alpha$$

Since $a_s = dv_s/dt = 0$, then

$$(0.5\text{ m/s})^2 + 0 = 2\cos 30°(0.6197\text{ rad/s})^2 + 2\sin 30°\alpha$$

$$\alpha = -0.415\text{ rad/s}^2 \qquad \textit{Ans.}$$

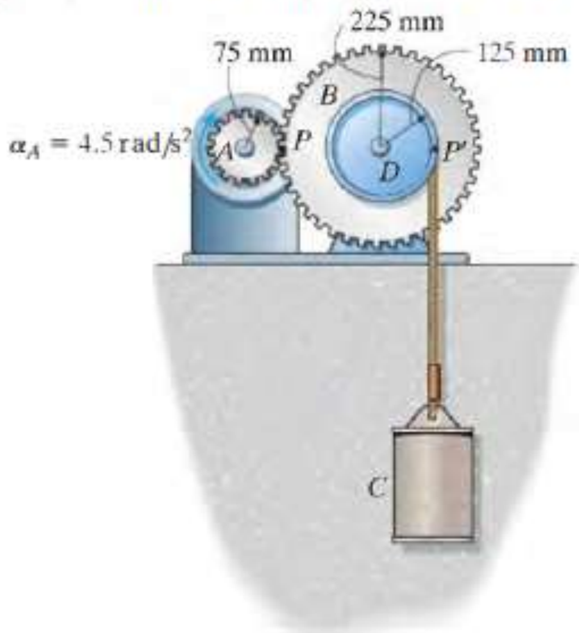
Because the result is negative, it indicates the window has an angular deceleration.

Problem Set 5.1

1.

F16–5. A wheel has an angular acceleration of $\alpha = (0.5\,\theta)\ \text{rad/s}^2$, where θ is in radians. Determine the magnitude of the velocity and acceleration of a point P located on its rim after the wheel has rotated 2 revolutions. The wheel has a radius of 0.2 m and starts from rest.

2. For a short period of time, the motor turns gear A with a constant angular acceleration of $\alpha_A = 4.5\ \text{rad/s}^2$, starting from rest. Determine the velocity of the cylinder and the distance it travels in three seconds. The cord is wrapped around pulley D which is rigidly attached to gear B.



3. Rod CD presses against AB, giving it an angular velocity. If the angular velocity of AB is maintained at $\omega = 5$ rad/s, determine the required magnitude of the velocity **v** of CD as a function of the angle θ of rod AB.